

Joe Chang

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As a GIS Analyst with three years at Google Maps, I've developed insightful skills in database/project management, quality control, and spatial analytics. Following the departure of my contract at Google Maps, I have continuously pursued my interest in coding and application development with Python and React to leverage GIS insights with growing technology and advancements. I am excited to bring my expertise and innovative approach to the next chapter of my GIS career.

GIS SKILLS

- ArcMap/Pro/StoryMaps/Online/QGIS
- Project/DB Management
- Arc FM/Survey123
- Technical Writing/Editing/Report
- Data Analysis/SQL/Macros
- Urban Planning
- CAD/As-Built/Sketchup
- QA/QC Workflow/Data
- Georeferencing/Rectifying Raster
- Cartography

CODING SKILLS

- Web GIS Development
- GIS API
- Python/ArcPy/GeoPandas
- ArcGIS Online
- React/JavaScript/NPM
- MongoDB/mongoose
- UI/UX Design
- Object Oriented Design
- Testing and Debugging
- Source Control Management

EXPERIENCE

GIS Analyst/Project Lead
Google Maps
Bothell, WA

November 2020 - August 2023

- Imported over 100+ Low Emission Zone features into Google Maps for major cities such as London, Paris, and Rome across EU nations
- Piloted project on Truck GPS covering over 1 million roads across the U.S. and Canada to ensure safe destination routing
- Collaborated with Google Engineers to improve and automate importing processes of large volume geo-spatial features into Google Maps
- Evaluated and deployed open data sources through ArcGIS to process datasets into a format that is feasible for the automation process
- Rigorous quality assurance methods through data validation procedures, spatial accuracy assessments, and data integrity to improve accuracy, effectively improving client satisfaction rating by 89%
- Documented technical procedures for the Team's workflow to standardize and improve the completion time, effectively improved TAT 140% from Q1 2023 – Q2 2023

PROJECTS

Google Maps: Deployed over 100 Low Emission Zones across major European cities. Each Low Emission Zone specifies critical information such as geometry of zone/boundary, time of operation, exceptions, and URL from official government source, and information provided in the official languages of the country.
<https://www.bbc.com/news/technology-58102651>

Collaborated with SE and GIS Team to create custom Python scripts validating attribute fields, effectively improving time of completion and accuracy. This script is used for various high-volume datasets specifically roads and addresses to flag potentially misspelled names and null/incorrect address numbers.

UW/GIS: A collaborative project with Forterra, using census tract data to breakdown the different socio-ethnic factors in correlation to food deserts across Snohomish County. Utilized Enterprise GDB to collaborate work and data simultaneously, our final delivery for this project is a deployed Web Map and Accessible Geodatabase for Forterra.

Per Scholas: Full-stack Development Course

React: <https://app.netlify.com/sites/roaring-boba-8ef40e/overview>

This project covers multiple components of react such as react-router-dom, and bootstrap to formulate an interactive NBA Application. Users can explore league-wide information via APIs and view specific team details using useState.

EDUCATION

Full-Stack Developer Boot Camp

November 2023

Per Scholas

16-week Full-Stack Software Engineering course that dives deep into the fundamental aspects of front and back-end software engineering: MERN stack, UI design, and System Architecture, and Back-End Development.

GIS Certificate

June 2021

University of Washington

Capstone project in collaboration with Forterra to create an interactive GIS Web Map, visualizing areas of food disparities across Snohomish County, WA and capsulating a written analysis report on the correlations based on census tract data.

Urban Planning B.S. minor: Geography

March 2020

Western Washington University

Condensed 10-week process simulating the roles and responsibilities of Urban Planners, our cohort led community meetings to gather initial feedback, which we then meticulously integrated into a mock comprehensive plan for Ferndale, WA. Our final deliverables included a detailed presentation and Comprehensive Plan brochure.